Computation Appendix to “Automation, Capital Taxation, and Public Insurance”

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C1 Computation and evidence: a reader's map

The paper uses calculations to clarify a mechanism, evaluate one fully specified rule, or diagnose a numerical route. It does not use a completed calculation as a substitute for an existence, feasibility, or optimality result. This map lets a reader start with a body use and recover the computed object, comparison, permitted inference, diagnostic, and record that must be consulted before extending it. The detailed modules give the equations, domains, and reconstruction materials; the main paper keeps only the qualification that changes the economic reading.

Paper use and comparison	Allowed inference	Decisive diagnostic and replication owner
Tax and capital propagation: deterministic interior characteristic and local capital–tax approximation around its named anchor	The tax condition and local propagation organize the mechanism; they do not give a global tax path or capacity frontier	Check the selected smooth interior branch, terminal condition, and relevant constraint slack. Owner: Brownian model specification and local-LQ computation record.
Portfolio mechanism: leading small-risk return demand, fiscal hedge, and payoff projection at checked Brownian nodes	A gross return motive and hedge can offset; a public claim removes only its spanned component	Check the component decomposition and checked-node squared-exposure span. Owner: Brownian small-risk calculation and its bound implementation record.
Order-two diagnostic: pointwise R3 policy correction compared with the cheaper R4 closure	R3 is exploratory method evidence; R4 shows why a shared low residual cannot validate a coefficient whose source derivatives are omitted	At the anchor, R4 reverses the R3 tax-speed sign and greatly enlarges the holding correction. Owner: Brownian local-methods record and frozen specification.
Fixed-rule incidence: announced automation transition versus no transition under common initial conditions, policy rule, and pricing	The imposed transition can have different worker and owner incidence under that rule; it is not a Ramsey gain	Check the paired transition and neutral-pricing accounts and valuations. Owner: fixed-policy OU specification and bound implementation record.
Public saving under the rule: a higher fund target or accumulation speed at otherwise fixed policy	Earlier transfers can be exchanged for later public resources; no optimal target is identified	Check consolidated public accounting, the retained domain, and stability. Owner: fixed-policy OU specification and bound implementation record.
Portfolio composition: safe-only versus domestic-equity rule at fixed saving and zero world price of risk	The available effect is unresolved at the reported accuracy; neither a ranking nor irrelevance follows	Reconcile the common continuation and uncertainty with the decision gap. Owner: fixed-policy OU numerical integration record.

Paper use and comparison	Allowed inference	Decisive diagnostic and replication owner
Nonzero-price-of-risk welfare: repriced equity continuation without replayed within-horizon public accounting	No welfare inference is allowed; the reported surfaces are excluded	Recompute wealth, transfers, consumption, and revaluation under the repriced process. Owner: fixed-policy OU numerical integration record.
Nonlinear Brownian continuation: candidate normalized HJB/PDE and its local bridge	Feasibility and method diagnostics only; no accepted nonlinear solution, capacity comparison, or welfare ranking	Require an independent evaluator, residual scaling, branch and boundary screens, and a reconstruction bundle. Owner: five-state Brownian PDE feasibility specification.
Event-arrival numerical attempt: candidate stationary and transition calculation	No numerical path, stationary point, tax magnitude, or welfare comparison is used	The current anchor and provenance failures quarantine the candidate; they do not establish nonexistence. Owner: event-arrival computation specification.

The table uses "diagnostic" in an economic sense. For a sign or ranking, the error assessment must be smaller than the decision gap after accounting, branch, boundary, and provenance checks have passed. A low algebraic residual on an approximation shared by two methods does not establish an independently recovered control; a simulated path inside a grid does not establish continuation solvency. Conversely, a rejected or quarantined comparison is retained because it describes the boundary of the present evidence rather than evidence for the opposite economic conclusion.

The replication route follows the model rather than the order in which code was written. For a Brownian mechanism statement, it begins with the selected deterministic state, the shock scale, the payoff convention, and the approximation order; it then checks the component decomposition, independent evaluation where required, and distances to the economic boundaries relevant to that statement. For a fixed-rule transition, it begins with the announced path and the complete tax, transfer, saving, ownership, and pricing rules; it then rebuilds the consolidated accounts and the worker and owner continuation evaluations on the same stochastic histories. The replication records retain detailed grids, source arrays, environments, and outputs. The appendix retains the smaller map needed to decide what a reported comparison means.

Two boundaries organize all of these calculations. First, fiscal value and an unconstrained portfolio direction are valuation objects, whereas public accounting, tax and transfer bounds, portfolio limits, and continuation solvency determine what can be implemented. A reference floor or numerical box may be a useful screen, but it is not the structural fiscal-capacity frontier. Second, the Brownian external claim and the installed domestic equity used in the event-arrival application have different ownership and pricing closures. Their numerical evidence is therefore not pooled.

C2 Visible outputs from the local benchmark

The main paper uses the local calculation to show the timing of worker consumption, tax adjustment, capital accumulation, and public finance. This section reproduces the two code outputs before presenting the numerical machinery that generated them. It is meant to let a reader see the economic object first and then decide which diagnostics matter.

Output identity and status. Both figures are copied without alteration from `implementation/lq-ramsey-numerics/outputs/dev-stage1-stage2a/figures/`. Their associated arrays and summaries are `irfs.csv`, `report.json`, and `summary.md` in that output bundle. The registered computation is `RUN-20260903T232308Z-CS003-R3-592e7816-01`; its current use ceiling is *exploratory only*. The figures may illustrate timing and local mechanisms, but they do not support a calibrated magnitude, a global solution, or a policy ranking.

C2.1 Four common-state transition experiments

Figure 1 shows productivity, automation, output-neutral, and claim-and-rental-base-neutral initial displacements propagated by the local system. Worker consumption is a flow deviation, the tax rate is a percentage-point deviation, capital is shown as a log deviation, and public net worth is a level deviation in the source normalization. The inherited capital stock and tax rate cannot jump. Their responses therefore begin at zero even when worker consumption and public net worth move at the impact date.

The output-neutral direction is a redistribution experiment, not a forecast. It offsets the output effect of the primitive innovations at impact while allowing wages, the rental base, and subsequent capital accumulation to move. The claim-and-rental-base neutral direction is a separate negative control: it removes two impact channels while leaving a worker-resource movement. These constructed directions reveal the local mechanism but need not remain neutral after the state propagates.

C2.2 Claim access and worker consumption

Figure 2 compares the same productivity and automation displacements with and without access to the external claim. The experiment changes the public asset menu while holding the inherited state and primitive innovation fixed. It therefore asks a well-defined question even though the current magnitude is not paper-ready.

The visible differences are small under this illustrative parameter vector. That is not a general null result. The current run has not supplied the matched nonlinear error, complete economic

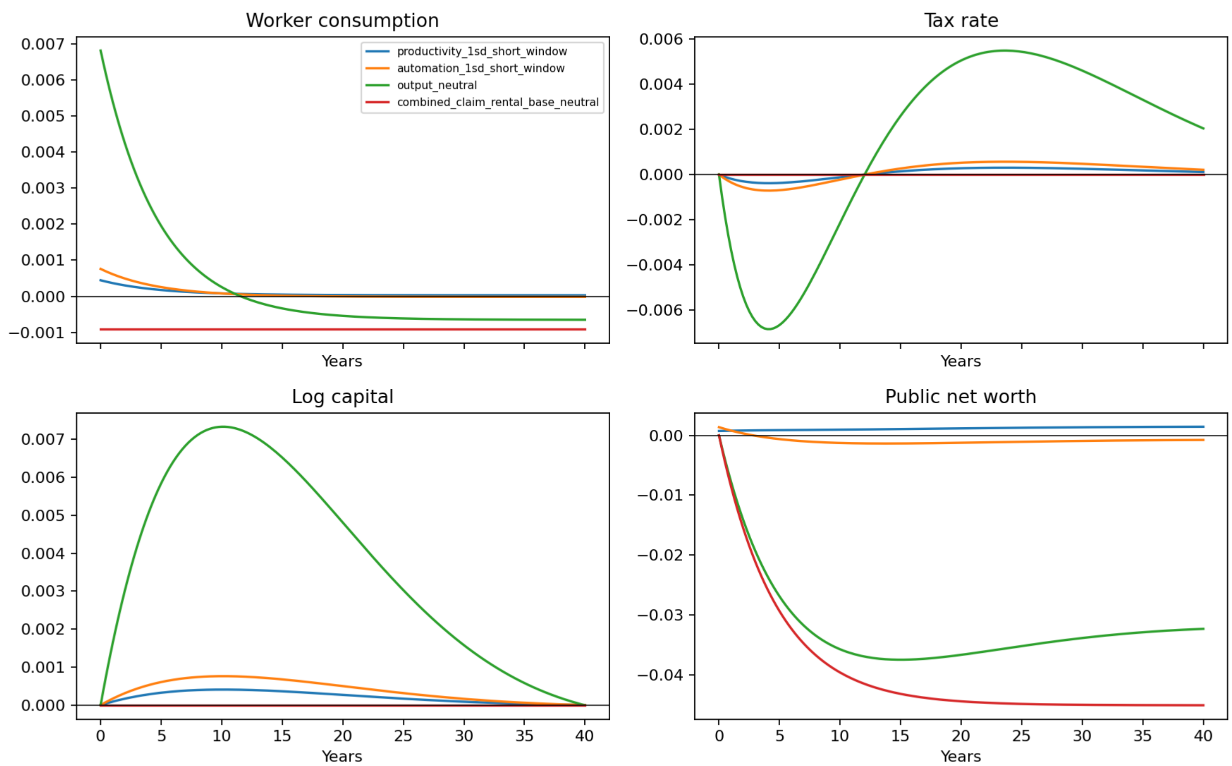

Figure 1: Development-stage local impulse responses. The figure is reproduced directly from the maintained output bundle. It is an illustrative mechanism calculation around a deterministic interior anchor, not a calibrated estimate.

boundary slacks, or an empirically disciplined shock scale. Its main use is diagnostic: it verifies that claim access is evaluated through worker consumption on a common path rather than through the gross public holding alone.

C2.3 What must be checked before quantitative use

A transition path becomes quantitative evidence only when the local and nonlinear calculations share primitives, inherited states, shock definitions, and policy restrictions. The discrepancy must be small relative to the consumption, tax, or welfare effect being discussed. Every displayed path must also report its distance from tax, transfer, portfolio, specialization, resource, and solvency boundaries. A Riccati residual establishes internal consistency of the local feedback; it does not establish that the original nonlinear economy stays on the same branch.

The order of the remaining computation appendix follows this evidentiary burden. It first identifies the permitted use of each calculation, then records the fixed-policy accounting benchmark, and finally gives the Brownian numerical methods and failure classification. Raw arrays and full run manifests remain in the replication record.

C3 Brownian local calculations

The Brownian calculations answer two limited questions. First, can a small net public position conceal a large return motive and a large fiscal hedge? Second, can a local quadratic approximation recover the policy correction needed for the paper's economic interpretation? The answer to the first is yes at the checked illustrative nodes. The second remains a method diagnosis rather than a quantitative result.

C3.1 Return demand, fiscal hedging, and span

The leading small-risk position is reported as two components, not only as a total. At the checked productivity node, compensated-return demand is approximately 12.70 and the fiscal-resource hedge is approximately -12.25, leaving a net position near 0.45. The maintained claim spans about 99.8 percent of squared raw fiscal exposure at the checked nodes. These values show why neither the net position nor the local spanned share is an adequate summary: the gross motives can be large even when their sum is small, and a small residual at selected nodes does not bound residual exposure elsewhere.

The calculation holds fixed the deterministic continuation, the common small-risk scaling, and the aligned payoff normalization used in the paper. It is an illustrative mechanism calculation. It does not identify a calibrated portfolio, establish feasible leverage, or convert the local projection into a fiscal-capacity or welfare result.

C3.2 The order-two policy diagnostic

The richer order-two calculation, denoted R3 in the computational records, evaluates corrections to tax speed and the total risky holding at named points on the deterministic path. At the anchor it gives

$$\nu^2 = -4.71141 \times 10^{-4}, \qquad s^2 = -7.72374 \times 10^{-4}. \tag{C3.1}$$

The diagnostic envelopes are 2.66 percent and 3.50 percent of the absolute values, respectively. At the productivity-impact node the pair is $(-4.27102 \times 10^{-4}, -7.49993 \times 10^{-4})$; at the automation-impact node it is $(-1.96936 \times 10^{-4}, -1.53355 \times 10^{-4})$. Each number is a pointwise correction, not an integrated policy path.

A cheaper closure, R4, demonstrates why a low matrix residual is not enough. At the same illustrative anchor it gives

$$\nu^2 = 0.00400364, \qquad s^2 = -0.315599. \tag{C3.2}$$

The tax-speed correction reverses sign, and the absolute risky-position correction is about 409 times the R3 value. The discrepancy arises because truncating the state before differentiation removes source derivatives that enter policy recovery. Since the two routes share the truncated object, agreement on its residual would not be an independent check of the policy coefficient.

The paper therefore uses R3 and R4 to choose and reject approximation routes. It does not interpret either pair as evidence that risk raises or lowers taxes, as a finite-risk portfolio response, or as a welfare comparison. A policy-facing calculation would also need an immutable specification, independent recovery of the relevant controls, perturbation-order scaling, and slack to the transfer, tax, specialization, position, and solvency boundaries.

C3.3 From a local calculation to a paper result

The decisive numerical error is the error in the economic quantity being reported. For a sign or ranking, the uncertainty envelope must remain well inside the distance to zero or to the competing value after accounting, branch, and boundary checks pass. A small equation residual cannot rescue a candidate that violates an identity or touches an economic constraint. The following modules state the full fixed-rule experiment, the independent Brownian evaluation route, and the nonlinear design needed for a wider domain. Executable grids, arrays, environments, and raw diagnostics remain with the replication records named in the opening map.

C4 A fixed-policy OU transition benchmark

This appendix describes a supplementary fixed-rule environment used to make one question concrete: given an announced automation transition and a pre-specified fiscal rule, how do factor incomes, public wealth, transfers, and group consumption move? It is not a re-solved Ramsey problem. It fixes aggregate capital, taxes, private domestic ownership, and the small-open pricing convention; it then evaluates a finite menu of fund-accumulation, transfer, and domestic-equity rules. The exercise is useful because it keeps the accounting of public ownership and foreign counterparties visible. Its scope is correspondingly narrow.

C4.1 The fixed rule and its environment

The state contains productivity and automation innovations following an OU process, an announced reference path for automation, and public financial wealth F. Aggregate capital $\bar{K}$ is fixed. Domestic private owners retain a fixed quantity K^K; the government may hold domestic equity K^G, subject to the available residual supply $\bar{K} - K^K$; and foreigners hold the remainder. This ownership convention makes a change in public domestic-equity exposure a trade with a residual foreign investor, not a confiscation of domestic owners' capital.

The inputs deliberately have different evidentiary roles. The announced automation path and reduced-form scale schedule define an incidence scenario. Persistence and volatility make the relevant risks visible. The safe rate and world prices of risk are financial assumptions of the small-open closure. Ownership shares and population weights map aggregate flows into groups. Fund targets, accumulation speeds, risky shares, drawdown caps, and transfer incidence are elements of a rule. Prices, returns, wealth paths, transfers, consumption, and consumption equivalents are outputs conditional on this entire packet. A factor-income path chosen to resemble a selected automation scenario is therefore not a validation target for the financial or policy blocks.

Production, factor incomes, and the announced path. The fixed-capital design abstracts from the capital-accumulation mechanism of the main Brownian Ramsey economy in order to isolate contemporaneous incidence and public balance sheet transmission. Output and factor payments are functions of the productivity and automation states and the fixed capital stock. The transition/no-transition comparison replaces the announced automation reference path by a constant mature value while holding the income, revenue, transfer, tax, and feedback anchors fixed. Common physical shocks are used in paired simulations. Hence a difference in the reported paths is a model implication of the imposed transition under the same rule, not evidence that a historical automation shock has been identified.

The reader should distinguish the tax base, domestic after-tax dividends, and worker income. A shift in factor income can increase a tax base while lowering worker income. That distinction is precisely why the model reports workers and domestic owners separately. It also prevents public equity income from being counted once as tax revenue and again as a dividend resource.

Private groups and ownership incidence. Workers receive their private labour income and the rule-determined worker transfer. Domestic owners receive private after-tax dividends, any owner transfer specified by the rule, and the value consequences of their fixed capital ownership. Foreign residual investors receive the after-tax dividends and capital gains on the residual quantity of domestic capital. They are necessary counterparties for the small-open pricing closure, but their welfare is not aggregated into a domestic social welfare number.

This separation yields a useful negative control. Conditional on the physical states, if owners receive no rule-dependent transfer and retain fixed private capital, changing the public saving or portfolio rule does not mechanically change their consumption. Any owner result then comes through the announced environment or a declared incidence channel, not through an unrecorded public acquisition. Conversely, an extension in which domestic owners sell capital to the government must record sale proceeds and the change in private financial wealth before it can be used for welfare analysis.

C4.2 Asset pricing and public accounting

Domestic equity is priced by an exogenous world stochastic discount factor. Let $Q^D(t,z,x)$ denote the price of a physical unit of domestic capital and $\widetilde{d}^D(t,z,x)$ its after-tax dividend. With OU generator $\mathcal{L}_t$, volatility matrix Σ, world safe rate r^w, and world price of risk λ^w, the pricing equation is

$$r^w Q^D = \widetilde{d}^D + \partial_t Q^D + \mathcal{L}_t Q^D - (\Sigma' \nabla Q^D)' \lambda^w. \tag{C4.1}$$

Equivalently, the price is the Feynman–Kac present value of future after-tax dividends under the world pricing measure, with the stated terminal continuation. The equation clarifies two otherwise easy confusions. First, changing λ^w reprices the claim; it does not change the physical OU law used for the transition. Second, a public position earns the claim's dividends and revaluation, not an arbitrary cash-flow series attached after the simulation.

The basic benchmarks are deliberately simple. With constant dividends and zero price of risk, $Q^D = \widetilde{d}^D / r^w$. On the zero-innovation announced path, the state-dependent price must coincide with the deterministic backward valuation. At selected on- and off-grid states, an independent Feynman–Kac calculation must agree within its Monte Carlo uncertainty. These checks are about the price object used in public accounting; they are not evidence that the world SDF is empirically correct.

Government accounting and the fair initial swap. Public financial wealth is allocated between a domestic-equity market value $Q^D K^G$ and a residual safe position. The safe asset earns the stated world safe rate. Taxes, transfers, equity dividends, equity revaluation, and saving flows are recorded at their specified times. The government cannot receive an initial equity claim for free. When a rule changes the equity share, the government exchanges domestic equity for the safe asset at Q_0^D while foreign residual investors take the opposite position. Date-zero public wealth, transfers, and domestic-owner resources consequently do not jump merely because the accounting label of the portfolio changes.

The consolidated identity is a useful audit rather than a decorative equation. It requires that domestic taxes and domestic transfers cancel in consolidation, while the foreign after-tax dividend flow, safe return, and public revaluation remain visible. It also requires that the tax component of a public capital claim not be counted again as a dividend. Pathwise budget and financial-account residuals are therefore reported alongside welfare calculations. A portfolio comparison that fails the fair-swap or consolidation check does not establish an insurance effect.

C4.3 Policy rules and stability

The benchmark evaluates rules rather than chooses them. A target-fund rule specifies a reference level $F^\star$ and an accumulation speed. A transfer rule allocates current resources between workers and owners according to named anchors and incidence weights. A portfolio rule specifies the fraction of public wealth invested in domestic equity, subject to nonnegative public wealth and the physical residual-ownership limit. The resulting experiment has four economically distinct comparisons.

The transition effect compares an announced automation transition with no transition under the same rule. The saving effect changes the target or accumulation speed at a fixed portfolio rule. The portfolio effect changes domestic-equity exposure relative to a safe-only rule at fixed saving. The scale–composition interaction asks whether a portfolio effect becomes relevant only after public wealth has accumulated. These comparisons hold fixed the initial state, taxes, income and transfer anchors, shock law, world pricing convention, ownership perimeter, and continuation method unless the changed object is explicitly named. Their finite menu has a best evaluated point, not an optimum.

Saving has an immediate accounting cost. On a slack reference path, extra primary saving reduces current transfers one for one before return effects. Later public resources can offset that sacrifice, but the date at which they do so and the group that receives them are consequences of the named transfer rule. This is why the benchmark reports transfer paths and wealth alongside consumption equivalents rather than leading with terminal fund size.

Mean and second-moment stability. A fund target can be mean reverting without being fiscally stable in a stronger sense. After linearization around a mature target, each rule has a mean transition matrix A_p. Mean stability requires A_p to be Hurwitz. A share rule, however, gives public wealth multiplicative return risk. In the isolated wealth direction, the second-moment condition takes the transparent form

$$2a_F + \omega_D^2 \|\sigma_D\|^2 < 0, \tag{C4.2}$$

where a_F is wealth-feedback drift, ω_D the domestic-equity share, and σ_D the equity-return loading. The full calculation checks the corresponding generalized covariance operator. An ordinary Lyapunov equation is exact only when the exposure is frozen; it can understate risk when exposure is proportional to wealth.

The distinction has direct reporting consequences. Stationary variance, infinite- horizon welfare, and tail claims require the second-moment condition and boundary slack, not merely a simulated mean path that returns toward target. A phase diagram should distinguish mean instability, second-moment instability, and contacts with transfer, wealth, or ownership bounds. A rule that hits a bound is not repaired by clipping the simulated position and then treated as an interior policy.

C4.4 Incidence, welfare, and evidence

Welfare is evaluated separately for workers and domestic owners under the fixed rule. The calculation combines simulated within-horizon consumption with a fixed-policy Markov continuation value. This continuation must use the same price, public-wealth rule, transfer rule, and ownership convention as the simulated path. Its Bellman residual, interpolation and boundary flags, and the distribution of terminal wealth relative to the retained continuation grid are part of the result's support.

The welfare objects should not be overread. They are not estimates, not a national Pareto criterion, and not an optimal-policy objective. They show how the stated rule allocates an illustrative transition between groups. Foreign residual investors are not silently made beneficiaries or payers. A consumption-equivalent number is reported only with its reference rule, horizon, terminal continuation convention, group, and underlying paths of transfers and private income.

Neutral-pricing evidence. The existing coarse neutral-price-of-risk surface supplies one limited result. For the transition/no-transition comparison under the named fixed rules, worker and owner consumption-equivalent signs are positive across the forty evaluated cells, with larger owner gains. The calculation is exploratory and illustrative. It is informative about incidence under the imposed automation path and transfer rule; it is not evidence that automation raises welfare generally, that either group is compensated in a different institution, or that the rule is Ramsey optimal.

The composition comparison is intentionally treated more cautiously. The frozen specification sets a decision threshold for neutral-price-of-risk portfolio effects, but the currently available reporting conventions and uncertainty envelopes are not reconciled sufficiently to support a common numerical bound or a signed welfare claim. The paper therefore records the comparison as unresolved at its current accuracy. This is not a statement that domestic equity is irrelevant. It says that, under the stated rule and neutral pricing, the existing calculation does not distinguish a meaningful composition effect from its numerical and reporting uncertainty.

Excluded repricing. The nonzero-price-of-risk welfare attempt is excluded. It repriced continuation objects but retained within-horizon public-wealth, transfer, consumption, and realized-revaluation accounting from the neutral-price simulation. Because repricing equity changes those flows, this combines different experiments. The defect is not repaired by a smaller tolerance or a new confidence interval: the welfare comparison is invalid until all accounting is replayed under the repriced process.

For the retained neutral-pricing comparisons, numerical evidence has a claim-shaped structure. Common shocks and identical anchors define the counterfactual. Pricing, ownership, budget, and fair-swap identities are structural gates. Exact OU moments, grid-boundary frequency, interpolation flags, independent prices, Monte Carlo error, horizon sensitivity, and transfer or wealth-boundary incidence delimit the reliable region. A comparison whose uncertainty interval crosses its sign, ranking, or economic threshold is labelled indistinguishable. A successfully executed program does not override a failed structural gate.

Reconstruction. Every paper figure or table drawn from this branch should identify the fixed-rule parameter packet; initial wealth and ownership state; announced transition; world-price convention; policy anchors; simulation and continuation conventions; code and environment identity; output artifact; and the diagnostic that supports the stated use. The replication material retains grids, seeds, raw paths, price and welfare evaluators, residual arrays, and hashes. This appendix retains the economic map: what changed, what remained fixed, what the calculation establishes, and which limitation prevents a stronger claim.

The fixed-policy benchmark thus has a clear role in the paper. It demonstrates how a public balance sheet transmits an announced automation transition under transparent rules and ownership accounting. It does not replace the Brownian Ramsey mechanism, provide a calibrated portfolio recommendation, or turn a price-taking financial closure into a general-equilibrium welfare result.

C5 Brownian computation: objects, diagnostics, and limits

This appendix explains how a candidate solution to the Brownian worker-welfare Ramsey problem is represented and checked. Its purpose is not to claim a solved global policy. The current computation is a feasibility and method programme: it asks whether a named smooth, interior continuation can be approximated on a declared central domain, recover economically meaningful controls, and survive independent residual and boundary checks. A small equation residual alone cannot establish global existence, uniqueness, optimality, or a natural fiscal-capacity frontier.

C5.1 Economic levels and solver coordinates

The reader-facing state is (z, x, K, N, τ): productivity, automation, installed capital, public financial net worth, and the inherited capital-income tax rate. Worker consumption c, the signed dollar risky position s, and tax speed $\nu = \dot{\tau}$ are controls. The computational representation normalizes these objects only after their economic roles are fixed. Log capital is used where $K > 0$ to stabilize scale; net worth is scaled by output or productive resources; and value and derivative representations are converted back to levels before budgets, controls, and constraints are reported. A normalized coordinate is never interpreted as a dollar debt limit.

The conversion is consequential. N is a financial stock, J is fiscal resource value for worker welfare, and $X = N + J$ is comprehensive worker-planner resources. The latter two include worker wages and therefore cannot be treated as Treasury collateral. Similarly, s is a signed dollar market value, not a portfolio share, and ν is a tax-speed control, not the tax rate. Every numerical table reconstructs transfers, tax receipts, adjustment costs, financial returns, and the safe residual in these economic units.

The normalized nonlinear HJB. On the maintained smooth full-specialization branch, a candidate value function solves the continuous-time HJB associated with worker log utility, the production law, the public budget, and exogenous world pricing. Before optimization, its essential structure is

$$\rho V = \max_{c,s,\nu} \{\log c + \mathcal{L}^{c,s,\nu} V\}, \tag{C5.1}$$

where the generator includes the two Brownian shocks, capital accumulation, public net worth, and tax adjustment costs. The solver uses a normalized equivalent of (C5.1); the independent evaluator rebuilds every term from level budgets, prices, and the transformation map. This separation is essential because a residual calculated from the same eliminated formulas or automatic-differentiation path as the solver is not independent evidence.

Interior first-order conditions recover the controls only on the declared concave, slack branch. Consumption is the inverse marginal value of public resources; the risky position trades expected excess return against the covariance derivative of the value; and tax speed trades the tax marginal value against its resource adjustment cost. If a transfer floor, tax bound, portfolio restriction, solvency boundary, or specialization boundary binds, these equalities give way to KKT or state-constraint conditions. A candidate recovered from the interior formulas must therefore be screened before it is called feasible.

C5.2 Domain, branches, and local benchmarks

The computational domain has three layers. The economic branch requires positive consumption, the maintained production specialization, admissible transfers and taxes, and conditional continuation solvency. A conservative reference floor is a compact screen used to measure distance from a known viable construction; it is not the unknown natural frontier. Finally, artificial box faces and control guardrails permit an approximation to be computed. Their slack must be reported separately. A path that avoids a box face but approaches an economic restriction is not an interior result; a path that avoids a reference floor does not establish global debt capacity.

The same distinction applies to branch selection. A nonlinear optimizer or neural representation can return a finite value on more than one numerical region. The maintained object is the branch connected to the specified deterministic continuation and satisfying the stated economic inequalities. Training loss, a first converged path, or a small collocation residual cannot select that branch by themselves.

The deterministic and LQ bridge. The deterministic characteristic is the first bridge between the nonlinear and local objects. It supplies a time-dependent continuation, terminal stable-tail condition, and fiscal-value derivatives. The local LQ system then linearizes around its named deterministic anchor. Its 2×2 Riccati block describes endogenous capital–tax feedback; the Sylvester block maps exogenous productivity and automation states into that feedback; and the discounted Lyapunov block accumulates quadratic welfare effects. This bridge has two uses. It provides warm starts and a low-dimensional benchmark for the nonlinear approximation, and it makes a falsifiable prediction: at small risk, the nonlinear value should agree with the deterministic plus order-two representation at the appropriate order.

The bridge is not an equivalence theorem. Strict additive-noise LQG has certainty- equivalent feedback; precautionary changes in tax speed or portfolio must come from the order-two expansion or a finite-risk calculation. Nor does a locally stable Riccati graph imply a stationary distribution for public wealth. The matched comparison is made only on common states, units, controls, and boundary conventions.

C5.3 Independent evaluation and numerical error

The independent evaluator accepts a frozen primitive and domain case, a candidate value and derivative representation, and recovered controls. It reconstructs the unoptimized and optimized HJB residuals, public and private accounting identities, FOCs or KKT conditions, price terms, and every maintained boundary distance at held-out states. It also evaluates manufactured values and deliberately perturbed inputs. These tests are designed to reveal sign errors, state-order mistakes, missing Itô terms, incorrect normalization, and optimizer/elimination mistakes that a training loss can miss.

Derivative recovery requires its own evidence. A policy correction depends on first and second derivatives of the value; differentiating through a solver's stopping logic is not a reliable independent route. The preferred checks compare derivatives obtained from the converged residual system with complete finite-difference re-solves and with the local analytic representation on the overlapping domain. A recovered policy is accepted only when its controls make independently evaluated FOC residuals small in economic units and remain inside the named interior region.

Residual scaling and the reliable region. Residuals are scaled by the economic quantity they protect. An HJB residual is paired with value and consumption units; a budget residual with flows or wealth; a portfolio FOC residual with the decision gap between a position and its relevant constraint. For the order-two expansion, the residual of $V^0 + \varepsilon^2 V^2$ must scale as ε^4 over the retained risk ladder. Halving ε should approach a factor of one sixteenth in the leading residual. Failure of this test diagnoses either an incorrect coefficient, a derivative error, or an expansion that has left its regular neighbourhood.

The reliable region is therefore reported as a set of states, risk scales, horizons, and active sets for which independent residuals, refinements, and slack checks agree. It is not the full training box. Any claimed sign, ranking, or boundary result requires an uncertainty envelope smaller than the predeclared decision gap. If the interval crosses zero, a ranking threshold, or a boundary, the output is indistinguishable or invalid rather than weak evidence in the desired direction.

Failure diagnosis and result use. Some failures are structural and cannot be averaged into a tolerance: a wrong specification, non-finite value, violated budget identity, failed KKT condition, unrecorded provenance, or breached economic boundary invalidates the affected output. Other failures localize the next diagnostic: mesh or horizon sensitivity calls for refinement; conditioning calls for a perturbation test; disagreement with the local benchmark calls for an overlap analysis. The computational record distinguishes exploratory output, diagnostic-only evidence, indistinguishable comparisons, and decision-resolved claims. Solver completion occupies none of these categories by itself.

The current nonlinear programme remains below the last category. It can support a careful statement about an interior candidate on a named central region only when its independent evaluator, branch checks, and boundary margins have passed for that region. It cannot support a global feedback, uniqueness, a natural debt limit, or a welfare ranking across unmatched regimes.

C5.4 Reconstructable output

A retained computational exhibit is backed by a compact bundle: the exact specification and fingerprint; primitive and parameter packet; transformation map; code and environment identity; candidate value and policy artifacts; independent residual arrays; held-out states; refinement and risk-ladder comparisons; boundary slack paths; and a plot-data table rebuilt independently from immutable output. The caption states the model branch, inherited state, comparison, units, reliable region, uncertainty envelope, and wording ceiling. The replication archive retains the full checkpoints, grids, seeds, logs, and hashes; the paper retains the economic route by which a reader can tell what the figure establishes and what it leaves open.

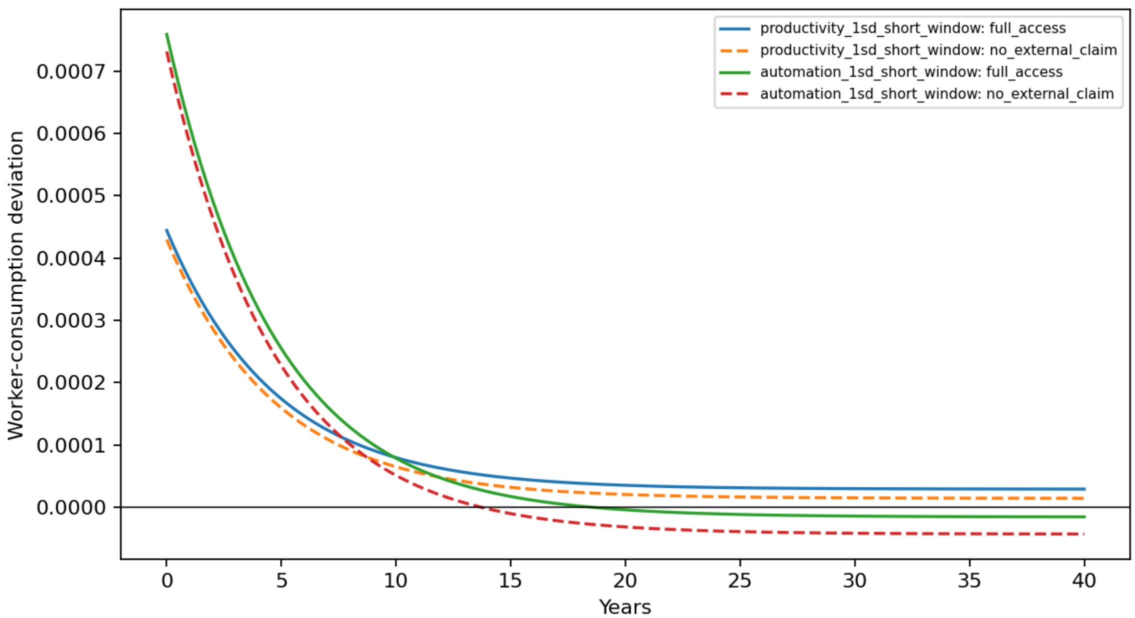

Figure 2: Worker-consumption responses with and without the external claim. Lines are direct development-stage outputs. Differences are not interpreted as calibrated welfare effects; the numerical and economic-domain requirements are given below.

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